MIDDLESEX SCHOOL | MATHEMATICS

Math 39 Course at a Glance

Precalculus with Data Analysis

Linear Modeling and Residuals

REQUIRED	
M39-LIN-001	I can calculate and interpret average rate of change with appropriate units.
M39-LIN-002	I can construct a linear function from two data points, a table, a graph, or a context.
M39-LIN-003	I can fit a linear regression model to data using technology and interpret its parameters.
M39-LIN-004	I can interpret a correlation coefficient as evidence about the direction and strength of a linear relationship.
M39-LIN-005	I can calculate residuals, construct a residual plot, and use its pattern to evaluate a model.
M39-LIN-006	I can use a linear model to make and interpret predictions, including solving for either variable.

Exponential Modeling

REQUIRED	
M39-EXP-001	I can distinguish linear and exponential change using rates, multipliers, tables, graphs, or contexts.
M39-EXP-002	I can construct an exponential function from two data points, a table, or a context.
M39-EXP-003	I can interpret the initial value, growth or decay factor, and growth or decay rate of an exponential model.
M39-EXP-004	I can determine and interpret doubling time or half-life from an exponential model.
M39-EXP-005	I can solve exponential and logarithmic equations in modeling contexts.
M39-EXP-006	I can fit an exponential regression model using technology and evaluate its fit.
M39-EXP-007	I can compare linear and exponential models and justify which is more appropriate for a data set or context.

Power Functions and Function Comparison

REQUIRED	
M39-POW-001	I can identify a power function and interpret the parameters in y = kx^p.
M39-POW-002	I can construct a power function from two data points or a proportional relationship.
M39-POW-003	I can relate the sign and size of k and p to the shape, domain behavior, and end behavior of a power function.
M39-POW-004	I can recognize and model direct, inverse, and other power-variation relationships.
M39-POW-005	I can solve equations involving integer or rational powers.
M39-POW-006	I can compare the long-run dominance of logarithmic, power, polynomial, and exponential functions.
M39-POW-007	I can fit a power regression model and interpret its parameters and correlation evidence.

Function Combinations and Polynomials

REQUIRED	
M39-POL-001	I can add, subtract, multiply, and divide functions represented by tables, graphs, or formulas.
M39-POL-002	I can determine where a combination of functions is positive, zero, or undefined.
M39-POL-003	I can determine polynomial end behavior from degree and leading coefficient.
M39-POL-004	I can identify zeros and multiplicities and relate them to polynomial graph behavior.
M39-POL-005	I can estimate a polynomial's possible degree from its zeros, turning points, inflection points, and end behavior.
M39-POL-006	I can construct a polynomial function from specified zeros, multiplicities, and a point.
M39-POL-007	I can match polynomial formulas and graphs using intercepts, multiplicity, degree, and end behavior.
M39-POL-008	I can build and optimize a polynomial model in a geometric or economic context.

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Precalculus with Data Analysis

Describing Data

REQUIRED

M39-STA-001	I can distinguish categorical and quantitative data.
M39-STA-002	I can construct and interpret frequency and relative-frequency tables.
M39-STA-003	I can construct and interpret bar charts, pie charts, and histograms using appropriate technology.
M39-STA-004	I can identify misleading features in a data display.
M39-STA-005	I can describe a distribution's modality, symmetry, skewness, and outliers.
M39-STA-006	I can calculate and interpret mean, median, and mode and select an appropriate measure of center.
M39-STA-007	I can calculate and interpret range, standard deviation, quartiles, and interquartile range.
M39-STA-008	I can construct and interpret a five-number summary and boxplot.
M39-STA-009	I can use percentiles and z-scores to locate and compare values within distributions.
M39-STA-010	I can compare distributions using their shape, center, spread, and unusual features.

Probability

REQUIRED

M39-PRO-001	I can organize categorical data in a contingency table or Venn diagram.
M39-PRO-002	I can calculate and interpret empirical, marginal, joint, and conditional probabilities from a contingency table.
M39-PRO-003	I can identify an experiment, outcomes, events, and a sample space.
M39-PRO-004	I can calculate theoretical probabilities when outcomes are equally likely.
M39-PRO-005	I can use complements to calculate probabilities.
M39-PRO-006	I can calculate probabilities of compound 'and' events and distinguish independent from dependent events.
M39-PRO-007	I can calculate probabilities of compound 'or' events and distinguish overlapping from disjoint events.
M39-PRO-008	I can compare experimental and theoretical probability and explain the law of large numbers.

Expected Value

EXTENSION

M39-EV-001	I can calculate the expected value of a discrete probability situation.
M39-EV-002	I can interpret expected value to evaluate a game, risk, or long-run decision.

Linearization of Nonlinear Data

EXTENSION

M39-POW-008	I can linearize power or exponential data and translate between the linearized and original models.